

## Letter of Intent and Previous Work

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My interests include mathematical modeling via convex and combinatorial optimization, reinforcement learning, probabilistic models, graph theory and dynamic programming algorithms. Interested in using probabilistic methods for creating suitable estimators and root cause analysis.

Currently researching various MCP-enabled RAG, Agentic, and GraphRAG architectures for text generation and their applicability in certain contexts relevant to Nike. Also working on a framework for evaluation of these workflows using built-in MLflow judges, standard evaluation metrics, and custom-tailored scorers and custom metrics.

Examples of my recent work and research interests are:

- a) RAG and Agentic AI workflows and relevant vector retrieval techniques as discussed in [this repo](#), [this repo](#) and [this repo](#); a framework for evaluation of these workflows using built-in MLflow judges, standard evaluation metrics, and custom-tailored scorers and custom metrics – for details see the sample code [here](#).
- b) Causal Inference realized via agentic workflows – for details see markdown documents and sample code [here](#).
- c) Actor-Critic design pattern and related algorithms applied to agentic workflows – for details see the sample code [here](#).
- d) Reinforcement learning Policy Gradient algorithms such as PPO, applying those to agentic workflows mentioned in a). For the purpose used the gymnasium environment and stable-baselines3 library. Ongoing work captured in [this document](#).
- e) Algorithms for Root Cause Analysis using Bayesian inference and Probabilistic Temporal Logic. For the purpose exploring the usability of causal inference algorithms included in the packages `causal-learn`, `causalml` and `dowhy`. For details refer to [this document](#) and [this document](#).
- f) Classical and Deep Learning algorithms for various image processing tasks such as synthetic noise generation, generative fill and shadow processing. Repos with code samples related to this work – [smooth gradient outpaint](#) and [image crop](#).
- g) Redesign of the online Fulfillment algorithm using the  $\varepsilon$ -constraint method, replacing the weighted objectives method for scalarization, reformulating the problem as Mixed Integer optimization problem. For details refer to [this document](#) and [this document](#).

My coding experience involve python, C++, C, Java.

Samples of my python code can be found here:

algorithms for smooth gradient outpainting

[https://github.com/dimitarpg13/smooth\\_gradient\\_outpaint](https://github.com/dimitarpg13/smooth_gradient_outpaint)

image crop algorithm using object segmentation with the Dichotomous Segmentation Deep Learning model:

[https://github.com/dimitarpg13/image\\_crop](https://github.com/dimitarpg13/image_crop)

Here are samples of my C++ code from past endeavors:

[https://github.com/google/or-](https://github.com/google/or-tools/compare/stable...dimitarpg13:ortools:dpg/PWL_solver_stable_py2.7_gtest_scipV6)

[tools/compare/stable...dimitarpg13:ortools:dpg/PWL\\_solver\\_stable\\_py2.7\\_gtest\\_scipV6](https://github.com/google/or-tools/compare/stable...dimitarpg13:ortools:dpg/PWL_solver_stable_py2.7_gtest_scipV6)

<https://github.com/dimitarpg13/testcode/blob/master/fraction.cpp>

[https://github.com/dimitarpg13/testcode/blob/master/fraction\\_mt.cpp](https://github.com/dimitarpg13/testcode/blob/master/fraction_mt.cpp)

[https://github.com/dimitarpg13/testcode/blob/master/fraction\\_bigint.cpp](https://github.com/dimitarpg13/testcode/blob/master/fraction_bigint.cpp)

[https://github.com/dimitarpg13/cpp\\_testcode/tree/master/SudokuQlik/src](https://github.com/dimitarpg13/cpp_testcode/tree/master/SudokuQlik/src)

And here are relevant documents to software design, architecture, coding techniques and design patterns:

<https://github.com/dimitarpg13/BigIndex/blob/main/PresentationDGueorguiev2018.pdf>  
[https://github.com/dimitarpg13/InsideTensorflow2Source/blob/master/Understanding\\_Tensorflow\\_2\\_source\\_code.pdf](https://github.com/dimitarpg13/InsideTensorflow2Source/blob/master/Understanding_Tensorflow_2_source_code.pdf)  
<https://github.com/dimitarpg13/UnderstandingPythonEcosystem>  
[https://github.com/dimitarpg13/inside\\_cpp\\_object\\_model](https://github.com/dimitarpg13/inside_cpp_object_model)

And here are few repos about C++ language details and features:

[https://github.com/dimitarpg13/cpp\\_effective\\_modern](https://github.com/dimitarpg13/cpp_effective_modern)  
[https://github.com/dimitarpg13/cpp\\_move\\_semantics](https://github.com/dimitarpg13/cpp_move_semantics)  
[https://github.com/dimitarpg13/cpp\\_templates\\_complete\\_guide](https://github.com/dimitarpg13/cpp_templates_complete_guide)  
[https://github.com/dimitarpg13/cpp\\_random\\_pieces](https://github.com/dimitarpg13/cpp_random_pieces)